

Statistics for Social Science Research

Lecture 1

August 5, 2026

Topics for today

- Introduction to the course
 - Let's get to know one another
 - Let's take a look at the course outline (separate handout)
- Introduction to statistics
 - Chapter 1 “Essentials of Statistics for the Behavioral Sciences”

(The slides in this power point are partly taken from the Power Point Lecture Slides to accompany the textbook “Essentials of Statistics for the Behavioral Sciences” by Frederick J Gravetter and Larry B. Wallnau.)

Chapter 1:

Introduction to Statistics

Learning Outcomes

1

- Know key statistical terms

2

- Know key measurement terms

3

- Know key research terms

4

- Know the place of statistics in science

5

- Understand summation notation

Math Skills Assessment

- Statistics requires basic math skills
- Inadequate basic math skills puts you **at risk** in this course
- Appendix A *Math Skills Assessment* helps you determine if you need a skills review
- Appendix A *Math Skills Review* provides a quick refresher course on those areas.
- The final *Math Skills Assessment* identifies your basic math skills competence

Differential Calculus Formulas

- $\frac{d}{dx} x^n = nx^{n-1}$
- $\frac{d}{dx} (fg) = fg' + gf'$
- $\frac{d}{dx} \left(\frac{f}{g} \right) = \frac{gf' - fg'}{g^2}$
- $\frac{d}{dx} f(g(x)) = f'(g(x))g'(x)$
- $\frac{d}{dx} (\sin x) = \cos x$
- $\frac{d}{dx} (\cos x) = -\sin x$
- $\frac{d}{dx} (\tan x) = \sec^2 x$
- $\frac{d}{dx} (\cot x) = -\csc^2 x$
- $\frac{d}{dx} (\sec x) = \sec x \tan x$
- $\frac{d}{dx} (\csc x) = -\csc x \cot x$
- $\frac{d}{dx} (e^x) = e^x$
- $\frac{d}{dx} (a^x) = a^x \ln a$
- $\frac{d}{dx} \ln x = \frac{1}{x}$
- $\frac{d}{dx} (\arcsin x) = \frac{1}{\sqrt{1-x^2}}$
- $\frac{d}{dx} (\arctan x) = \frac{1}{1+x^2}$

1.1 Statistics, Science and Observations

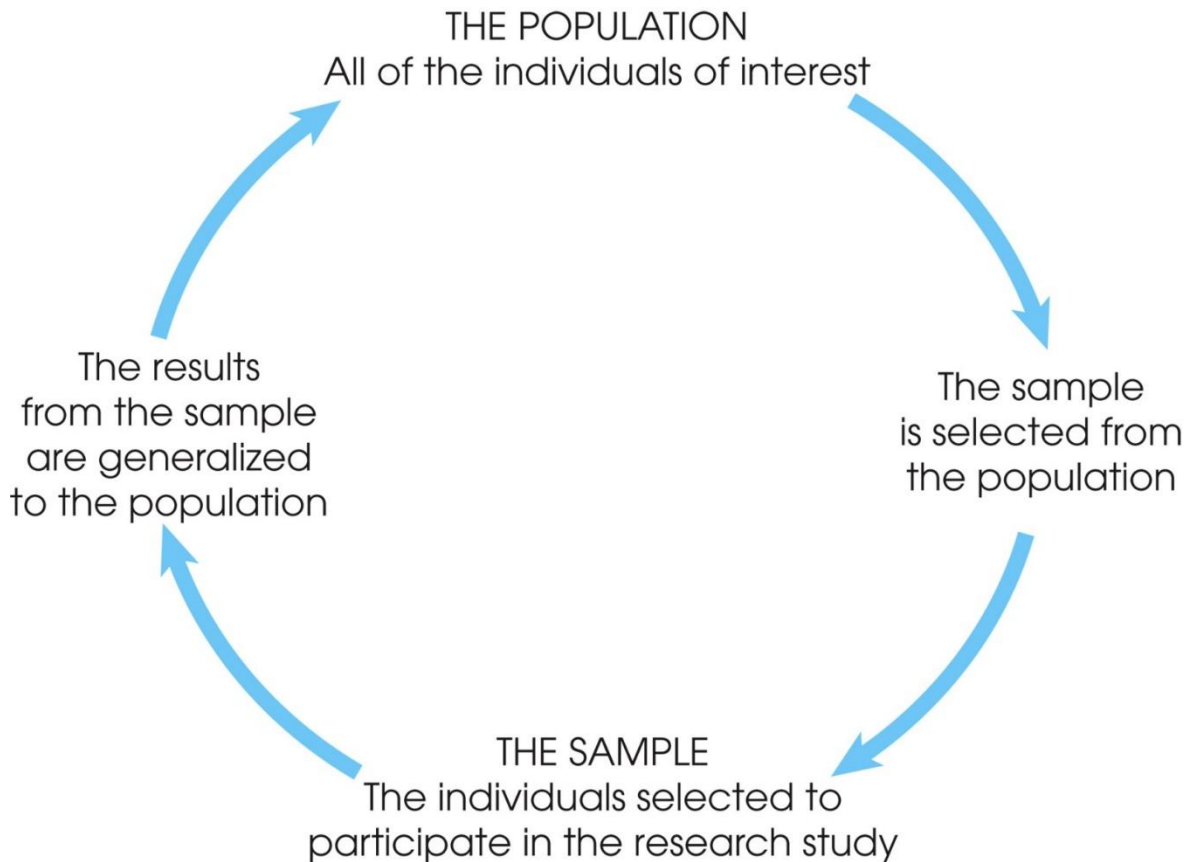
- “Statistics” means “*statistical procedures*”
- Uses of Statistics
 - Organize and summarize information
 - Determine exactly what conclusions are justified based on the results that were obtained
- Goals of statistical procedures
 - Accurate and meaningful interpretation
 - Provide standardized evaluation procedures

1.2 Populations and Samples

- Population
 - The set of all the individuals of interest in a particular study
 - Vary in size; often quite large
- Sample
 - A set of individuals selected from a population
 - Usually intended to represent the population in a research study

Figure 1.1

Relationship between population and sample

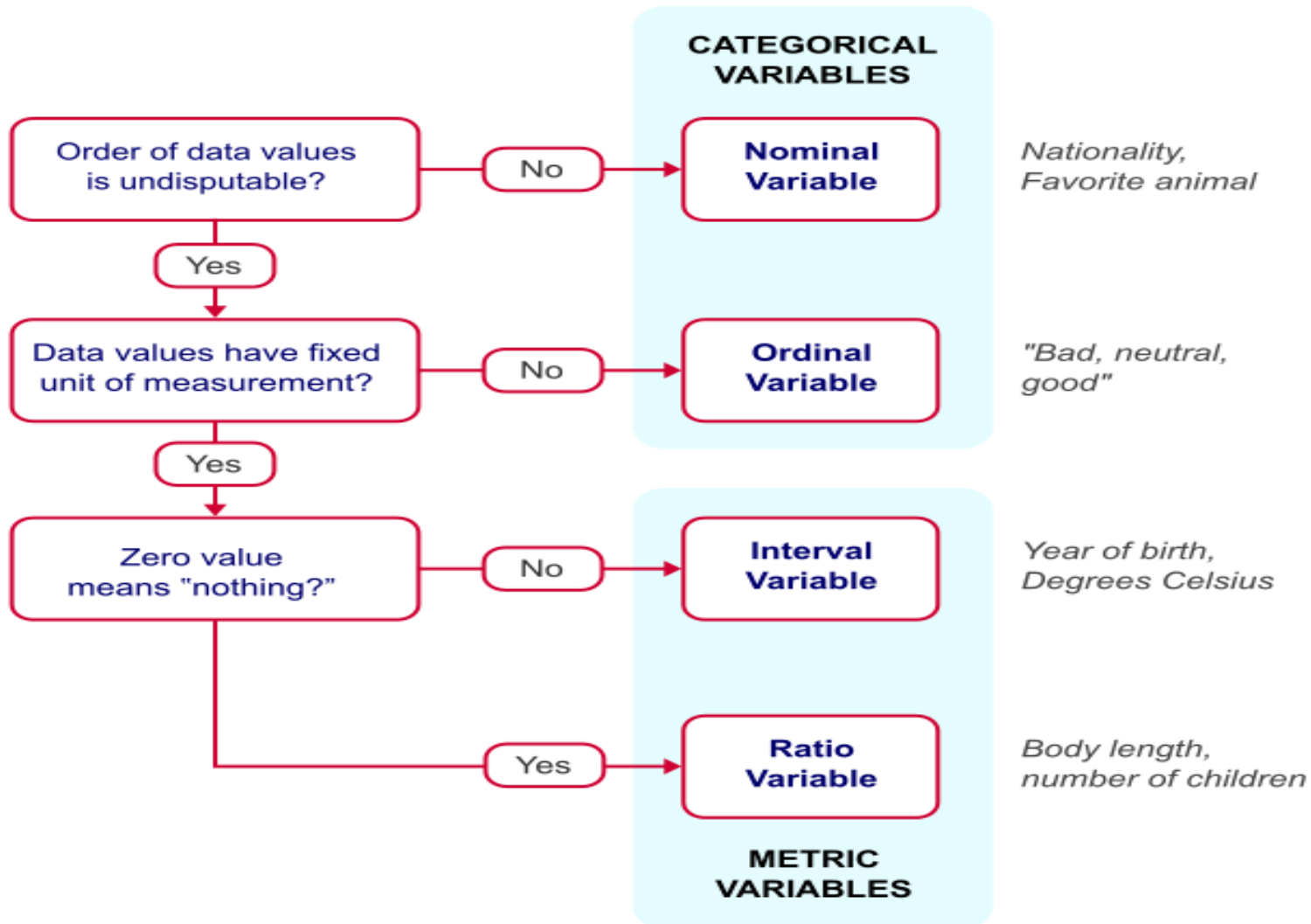


Variables and Data

- **Variable**
 - Characteristic or condition that changes or has different values for different individuals
- **Data (plural)**
 - Measurements or observations of a variable
- **Data set**
 - A collection of measurements or observations
- **A datum (singular)**
 - A single measurement or observation
 - Commonly called a **score** or **raw score**

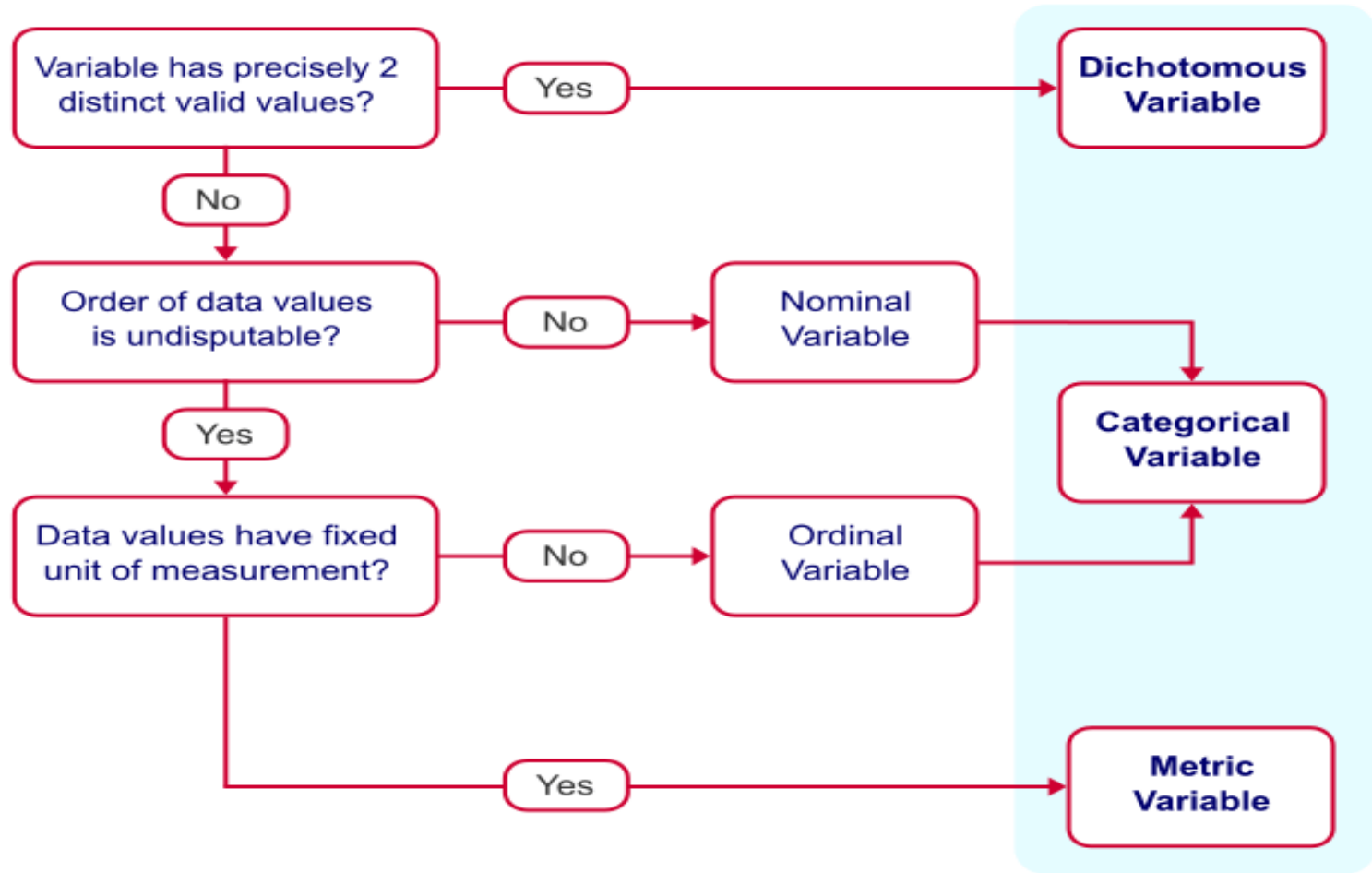
Variables and Data

MEASUREMENT LEVELS - CLASSICAL APPROACH



Variables and Data

MEASUREMENT LEVELS - MODERN APPROACH



Parameters and Statistics

- **Parameter**

- A value, usually a numerical value, that describes a population
- Derived from measurements of the individuals in the population

- **Statistic**

- A value, usually a numerical value, that describes a sample
- Derived from measurements of the individuals in the sample

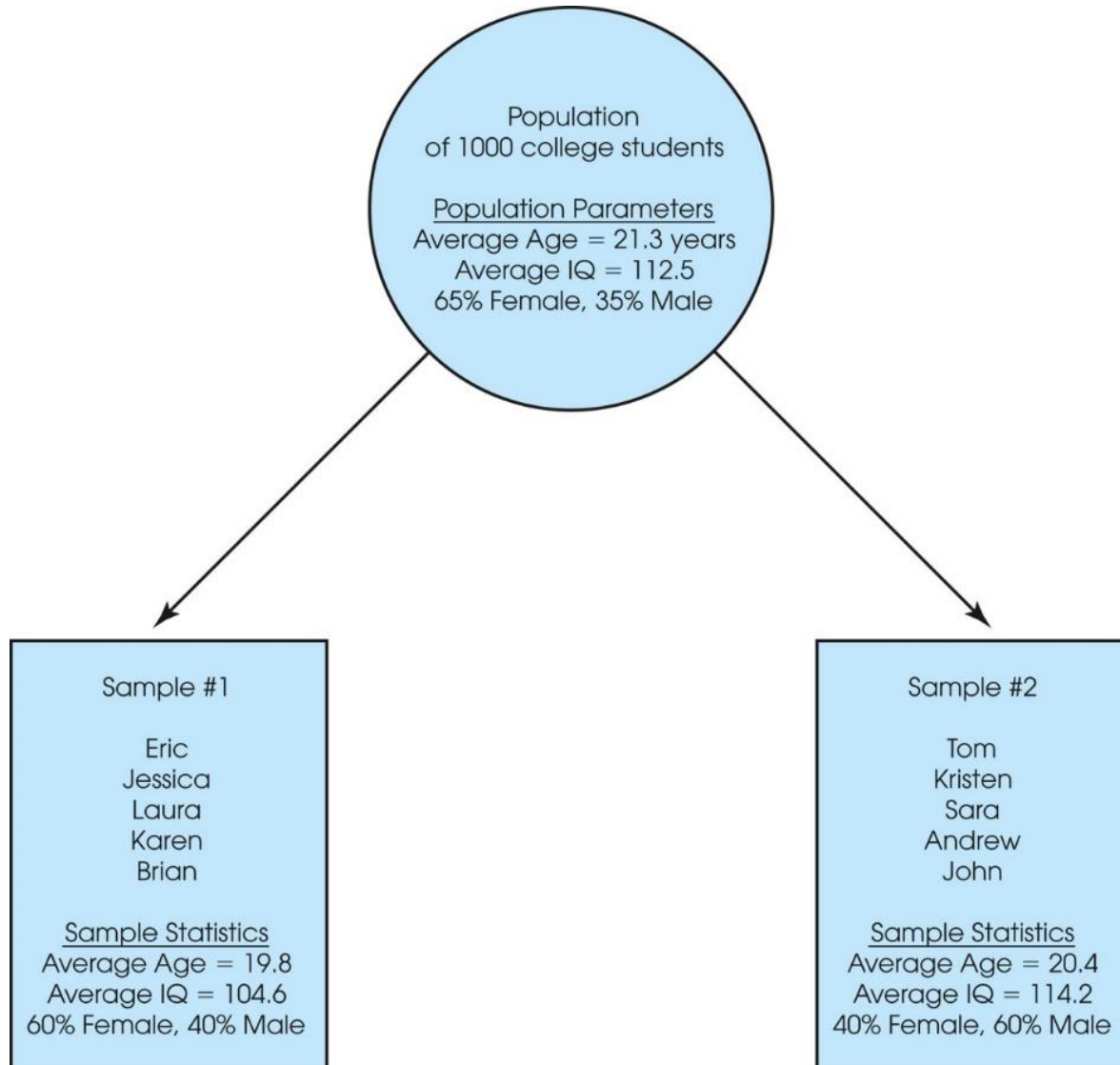
Descriptive & Inferential Statistics

- **Descriptive statistics**
 - Summarize data
 - Organize data
 - Simplify data
- Familiar examples
 - Tables
 - Graphs
 - Averages
- **Inferential statistics**
 - Study samples to make generalizations about the population
 - Interpret experimental data
- Common terminology
 - “Margin of error”
 - “Statistically significant”

Sampling Error

- Sample is never identical to population
- Sampling Error
 - The discrepancy, or amount of error, that exists between a sample statistic and the corresponding population parameter
- Example: Margin of Error in Polls
 - “This poll was taken from a sample of registered voters and has a margin of error of plus-or-minus 4 percentage points” (Box 1.1)

Figure 1.2
A demonstration of sampling error



1.3 Data Structures, Research Methods, and Statistics

- Individual Variables
 - A variable is observed
 - “Statistics” describe the observed variable
 - Category and/or numerical variables
- Relationships between variables
 - Two variables observed and measured
 - One of two possible data structures used to determine what type of relationship exists

Relationships Between Variables

- Data Structure I: The Correlational Method
 - One group of participants
 - Measurement of two variables for each participant
 - The goal is to describe the type and magnitude of the relationship
 - Patterns in the data reveal relationships
 - Non-experimental method of study

Figure 1.4

Data structures for studies evaluating the relationship between variables

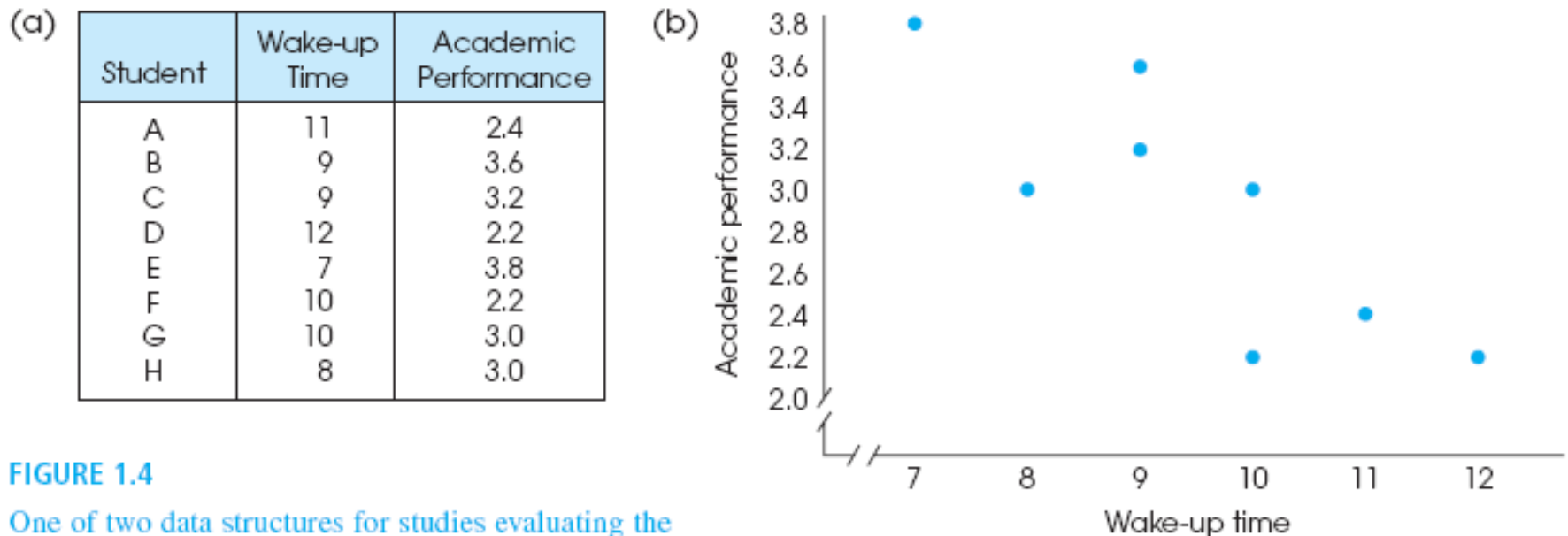


FIGURE 1.4

One of two data structures for studies evaluating the relationship between variables. Note that there are two separate measurements for each individual (wake-up time and academic performance). The same scores are shown in a table (a) and a graph (b).

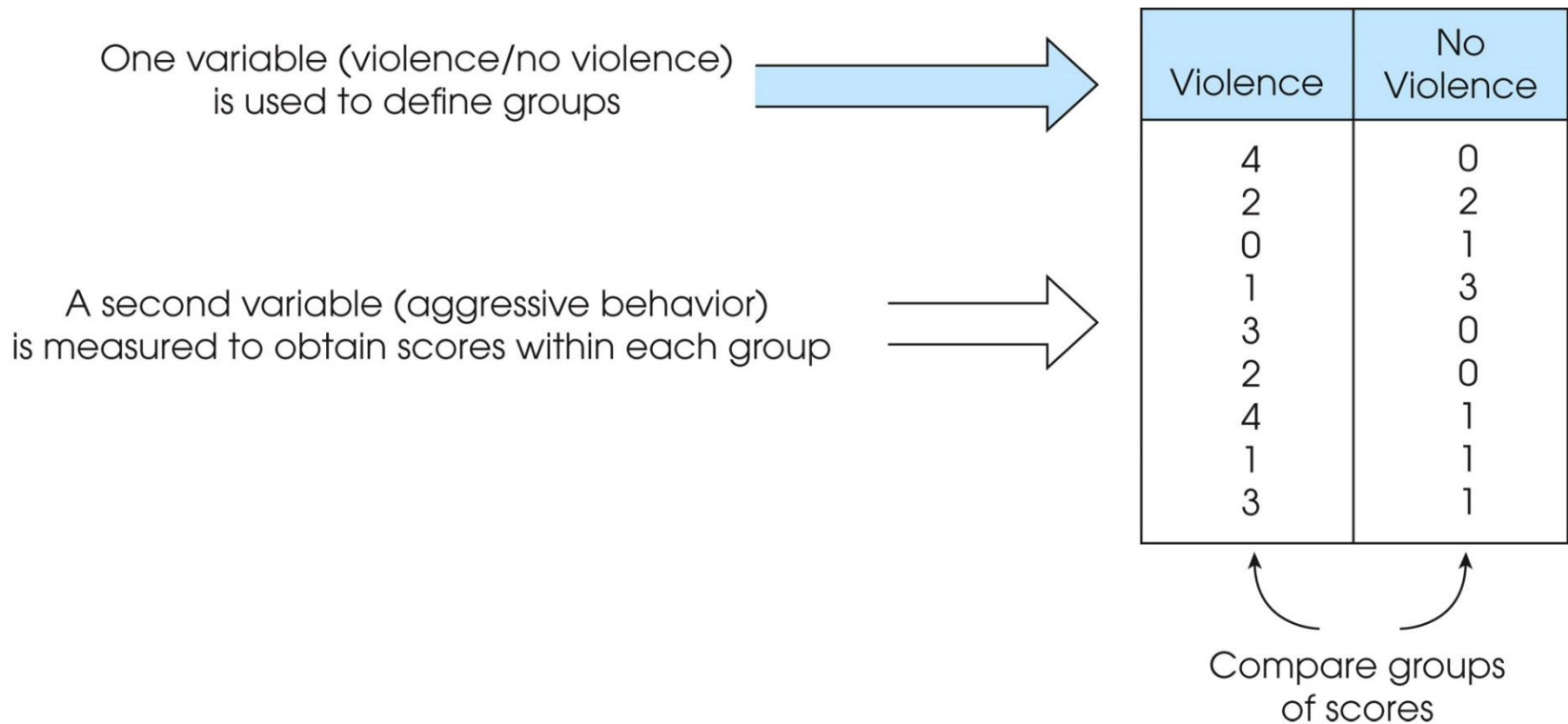
Correlational Method Limitations

- Can demonstrate the existence of a relationship
- Does ***not*** provide an explanation for the relationship
- Most importantly, does ***not*** demonstrate a cause-and-effect relationship between the two variables

Relationships Between Variables

- Data Structure II: Comparing two (or more) groups of Scores
 - One variable defines the groups
 - Scores are measured on second variable
 - Both experimental and non-experimental studies use this structure

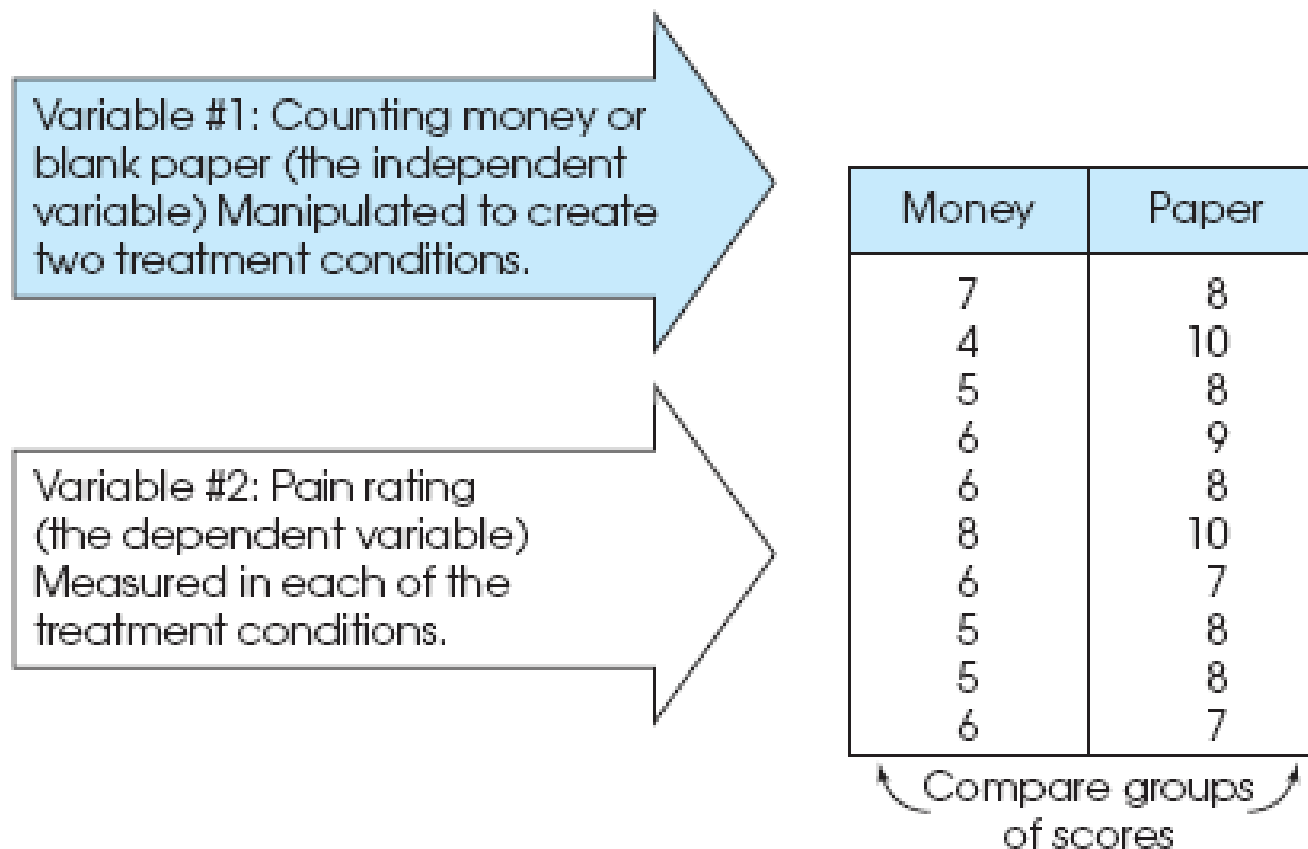
Figure 1.5
Data structure for studies comparing groups



Experimental Method

- **Goal of Experimental Method**
 - To demonstrate a cause-and-effect relationship
- **Manipulation**
 - The level of one variable is determined by the experimenter
- **Control** rules out influence of other variables
 - Participant variables
 - Environmental variables

Figure 1.6
The structure of an experiment



Independent/Dependent Variables

- **Independent Variable** is the variable manipulated by the researcher
 - Independent because no other variable in the study influences its value
- **Dependent Variable** is the one observed to assess the effect of treatment
 - Dependent because its value is thought to depend on the value of the independent variable

Experimental Method: Control

- **Methods of control**
 - Random assignment of subjects
 - Matching of subjects
 - Holding level of some potentially influential variables constant
- **Control condition**
 - Individuals do not receive the experimental treatment.
 - They either receive no treatment or they receive a neutral, placebo treatment
 - Purpose: to provide a baseline for comparison with the experimental condition
- **Experimental condition**
 - Individuals do receive the experimental treatment

Non-experimental Methods

- Non-equivalent Groups
 - Researcher compares groups
 - Researcher cannot control who goes into which group
- Pre-test / Post-test
 - Individuals measured at two points in time
 - Researcher cannot control influence of the passage of time
- Independent variable is **quasi-independent**

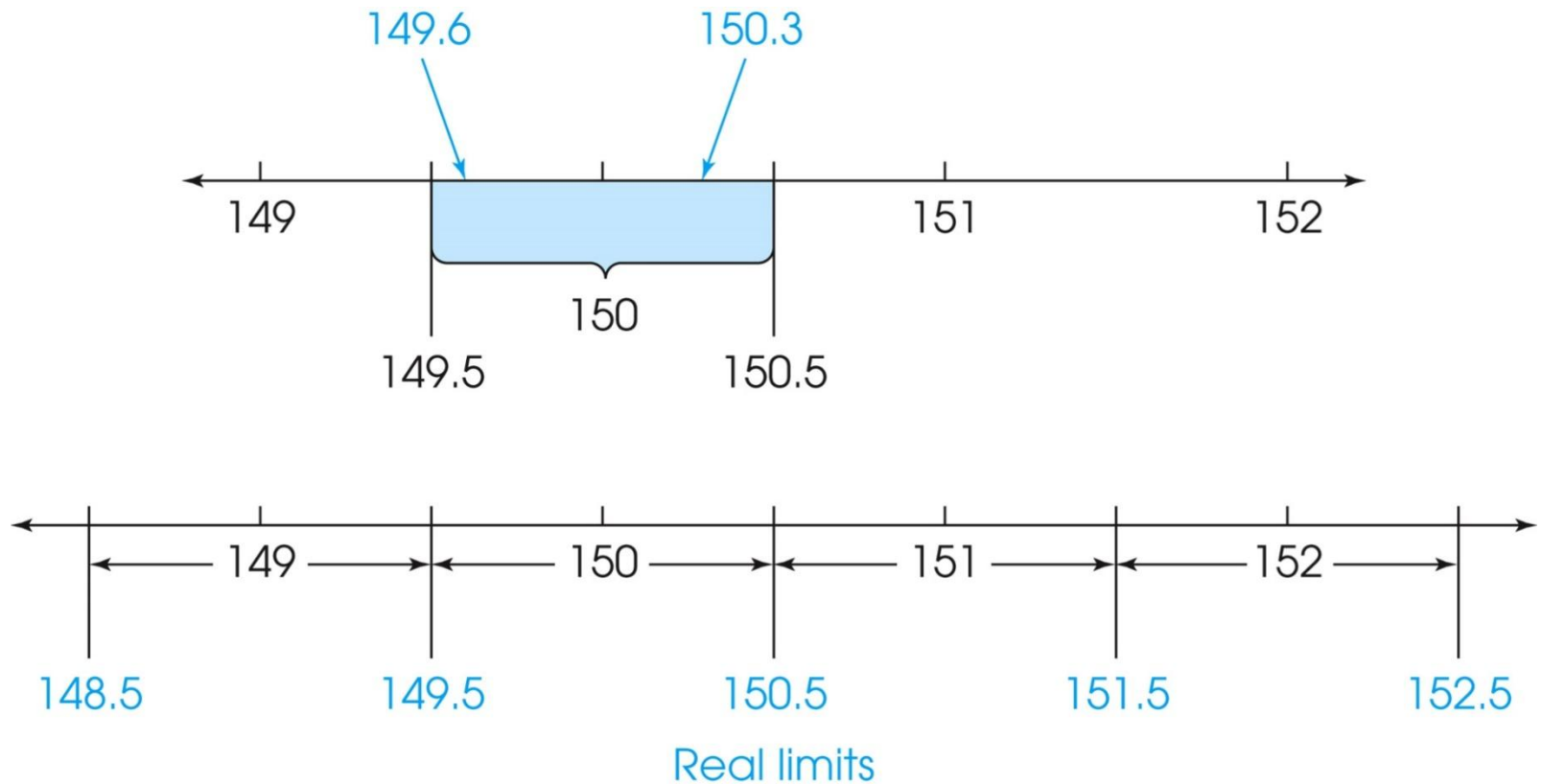
1.4 Variables and Measurement

- Scores are obtained by observing and measuring variables that scientists use to help define and explain external behaviors
- The process of measurement consists of applying carefully defined measurement procedures for each variable

Discrete and Continuous Variables

- **Discrete variable**
 - Has separate, indivisible categories
 - No values can exist between two neighboring categories
- **Continuous variable**
 - Have an infinite number of possible values between any two observed values
 - Every interval is divisible into an infinite number of equal parts

Figure 1.8
Example: Continuous Measurement



Real Limits of Continuous Variables

- **Real Limits** are the boundaries of each interval representing scores measured on a continuous number line
 - The real limit separating two adjacent scores is exactly halfway between the two scores
 - Each score has two real limits
 - The upper real limit marks the top of the interval
 - The lower real limit marks the bottom of the interval

Scales of Measurement

- Measurement assigns individuals or events to categories
 - The categories can simply be names such as male/female or employed/unemployed
 - They can be numerical values such as 68 inches or 175 pounds
- The complete set of categories makes up a scale of measurement
- Relationships between the categories determine different types of scales

Scale	Characteristics	Examples
Nominal	•Label and categorize •No quantitative distinctions	•Gender •Diagnosis •Experimental or Control
Ordinal	•Categorizes observations •Categories organized by size or magnitude	•Rank in class •Clothing sizes (S,M,L,XL) •Olympic medals
Interval	•Ordered categories •Interval between categories of equal size •Arbitrary or absent zero point	•Temperature •IQ •Golf scores (above/below par)
Ratio	•Ordered categories •Equal interval between categories •Absolute zero point	•Number of correct answers •Time to complete task •Gain in height since last year

1.5 Statistical Notation

- Statistics uses operations and notation you have already learned
 - Appendix A has a Mathematical Review
- Statistics also uses some specific notation
 - Scores are referred to as X (and Y)
 - N is the number of scores in a population
 - n is the number of scores in a sample

Summation Notation

- Many statistical procedures sum (add up) a set of scores
- The summation sign Σ stands for summation
 - The Σ is followed by a symbol or equation that defines what is to be summed
 - Summation is done after operations in parentheses, squaring, and multiplication or division.
 - Summation is done before other addition or subtraction